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Aryabhatta Knowledge University, Patna

B.Tech 6th Semester Examination, 2022 (New Course)

Time: 03 Hours

Code: 110601 ***Subject: Measurements and Instrumentation*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven of the following questions: **[14]**

(a) Explain the temperature and friction errors of the measuring instruments.

(b) Discuss any three errors of permanent magnet moving-coil instruments.

(c) What are the causes of errors in current transformers?

(d) Write the limitations of a Wheatstone bridge.

(e) Explain general working principle of a potentiometer.

(f) Explain eddy current and hysteresis losses.

(g) Explain the transfer characteristics of digital to analog conversion.

(h) What is a low power factor electro-dynamometer type wattmeter?

(i) Explain the displacement constant and the constant of inertia for a galvanometer.

(j) Write the advantages and precautions to be considered for a Wheatstone bridge.

Q.2 Solve both questions: **[14]**

(a) Describe the methods used for measurement of low pressures using (i) thermocouple vacuum gauge; (ii) pirani gauge; (iii) ionization-type vacuum gauge.

(b) Discuss the working of strain gauge-type of torque transducers. Explain its advantages and disadvantages.

Q.3 Solve both questions: **[14]**

(a) Write the working principle of attraction-type moving-iron instruments using schematic diagrams.

(b) Describe the construction and working principle of (i) infrared pyrometers and (ii) disappearing filament optical pyrometers. Explain their applications, temperature range, advantages and disadvantages.

Q.4 Solve both questions: **[14]**

(a) Discuss standardization, applications and precautions for a slide wire d.c. potentiometer.

(b) A galvanometer has the inertia constant 1.8, damping constant 6.5, deflection constant 8500N−m/*ampere*. Determine the value of the restoring constant that would give critical damping. Under these conditions what will be the transient deflection when a current of 1.2 milliamperes is being measured? What will be the deflection at the end of 3 seconds?

(c) Discuss vibration galvanometer using sufficient schematic diagrams and give its general features.

Q.5 Solve both questions: **[14]**

(a) Describe the method of measurement of differential pressure using an inductive transducer.

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(b) A barium titanate piezoelectric pickup has dimensions of 5mm×5mm×1.5mm and a voltage sensitivity of 0.015V m/N. Relative permittivity of the barium titanate is 1200 and modulus of elasticity is $10\times10^{10}N/m^2$. Determine (i) output voltage, (ii) charge sensitivity, (iii) charge generated and the capacitance of the pickup. The force applied to the pickup is 12 N.

(c) Explain the theory of radiation pyrometers. Describe the different radiation receiving elements.

Q.6 Solve all three questions: **[14]**

(a) Draw the circuit of a Kelvin's double bridge used for measurement of low resistances. Derive the condition for balance.

(b) What are the different factors which affect the precision measurement of medium resistances with Wheatstone bridge? Explain how their effects are minimized/eliminated.

Q.7 Solve both questions: **[14]**

(a) Write the working principle of megger along with a suitable block diagram.

(b) A capacitive transducer is used in pressure measuring instrument which has a spacing of 3.8 mm between its diaphragms. A pressure of $500kN/m^2$ produces average deflection of 0.25 mm of diaphragm of transducer. A transducer which has a capacitance of 280 pF before application of pressure is connected to oscillator circuit having a frequency 100 kHz. Find the change in frequency of oscillator after the application of pressure to the transducer.

Q.8 Solve both questions: **[14]**

(a) Explain the working principle of a digital storage oscilloscope using suitable block diagram. Write the advantages of digital storage oscilloscope.

(b) Write the working principle of digital multimeter along with a suitable block diagram.

Q.9 Solve both questions: **[14]**

(a) Give the instrumentation diagram of an ECG monitoring system and explain in detail.

(b) What is EEG and what are the different types of leads used in EEG? Write the differences between time domain EEG curve and EEG histogram.

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